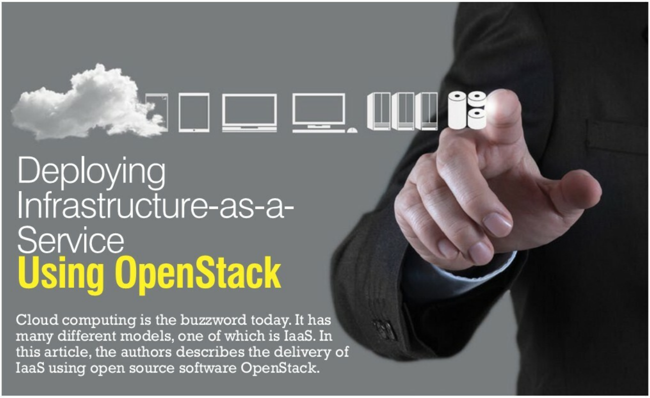

Deploying Infrastructure-as-a-Service Using OpenStack

Cloud computing is the buzzword today. It has many different models, one of which is IaaS. In this article, the authors describes the delivery of IaaS using open source software OpenStack.

Nowadays, cloud computing has become mainstream both in the research as well as corporate communities. A number of cloud services providers offer computing resources in different domains as well as forms. Cloud computing refers to the delivery of computing resources as a service rather than as a product. In cloud services, the computing power, devices, resources, software and information are delivered to clients as utilities. Classically, such services are provided and transmitted by using network infrastructure or simply delivered over the Internet.

Infrastructure as-a-service (IaaS)

IaaS includes the delivery of computing infrastructure such as a virtual machine, disk image library, raw block storage, object storage, firewalls, load balancers, IP addresses, virtual local area networks and other features on-demand from a large pool of resources installed in data centres. Cloud providers bill for the IaaS services on a utility computing basis; the cost is based on the amount of resources allocated and consumed.

OpenStack: a free and open source cloud computing platform

OpenStack is a free and open source, cloud computing software platform that is widely used in the deployment of infrastructure-as-a-Service (IaaS) solutions. The core technology with OpenStack comprises a set of interrelated projects that control the overall layers of processing, storage and networking resources through a data centre that is managed by the users using a Web-based dashboard, command-line tools, or by using the RESTful API.

Currently, OpenStack is maintained by the OpenStack Foundation, which is a non-profit corporate organisation established in September 2012 to promote OpenStack software as well as its community. Many corporate giants have joined the project, including GoDaddy, Hewlett Packard, IBM, Intel, Mellanox, Mirantis, NEC, NetApp, Nexenta, Oracle, Red Hat, SUSE Linux, VMware, Arista Networks, AT&T, AMD, Avaya, Canonical, Cisco, Dell, EMC, Ericsson, Yahoo!, etc.

OpenStack users

• AT&T	• Purdue University
• Stockholm University	• Red Hat
• SUSE	• CERN
• Deutsche Telekom	• HP Converged Cloud
• HP Public Cloud	• Intel
• KT (formerly Korea Telecom)	• NASA
• NSA	• PayPal
• Disney	• Sony
• Rackspace Cloud	• SUSE Cloud Solution
• Wikimedia Labs	• Yahoo!
• Walmart	• Opera Software